CLINICAL AI CONTROL PLANE

Architecture Diagrams

CAG-02 · System Context and Request Flow

Prepared by Engineering · 2026-08-21

Clinical AI Platform Group

PHI PROTECTION · FULL AUDIT TRAIL · HIPAA ALIGNED

Clinical AI Control Plane: Architecture Diagrams

System context · Governed request flow · State machine · Deployment topology

SECTION 01

System context

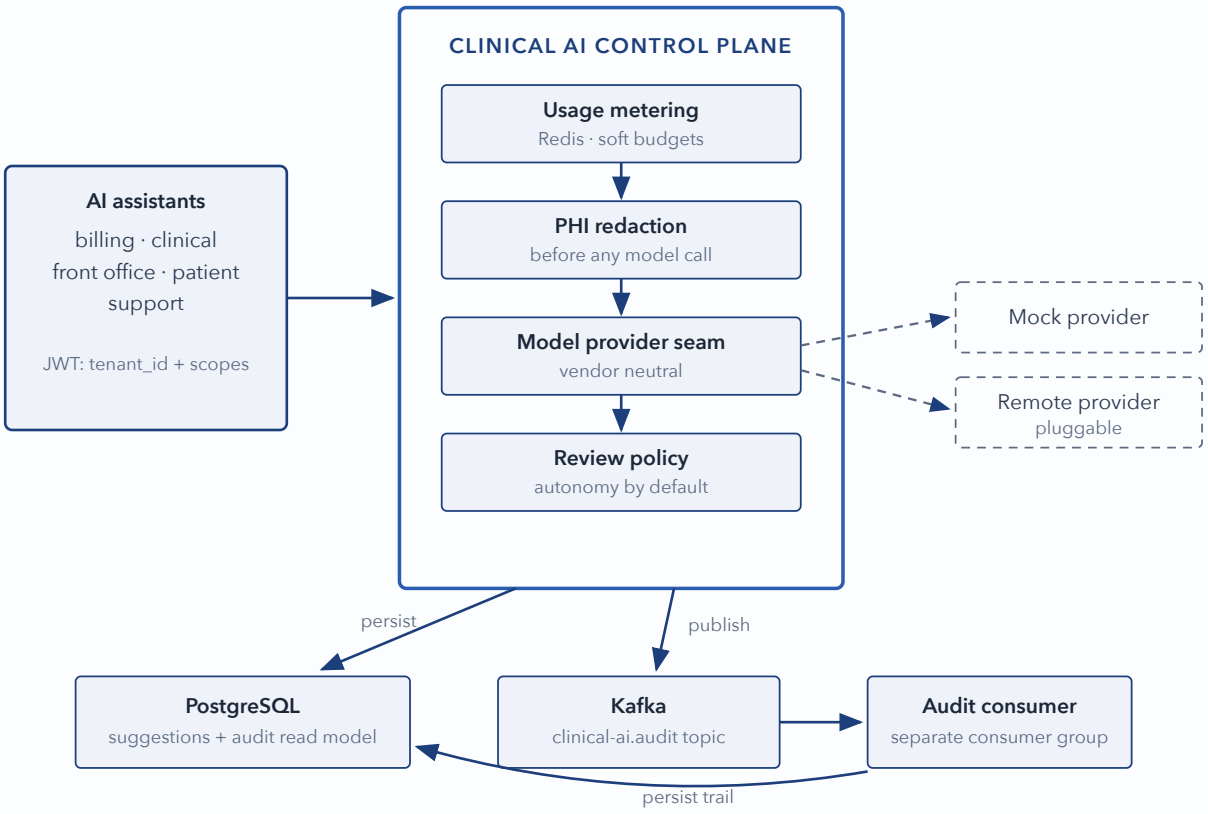


Figure 1 · The control plane between the assistant fleet and the model layer

SECTION 02

Governed request flow

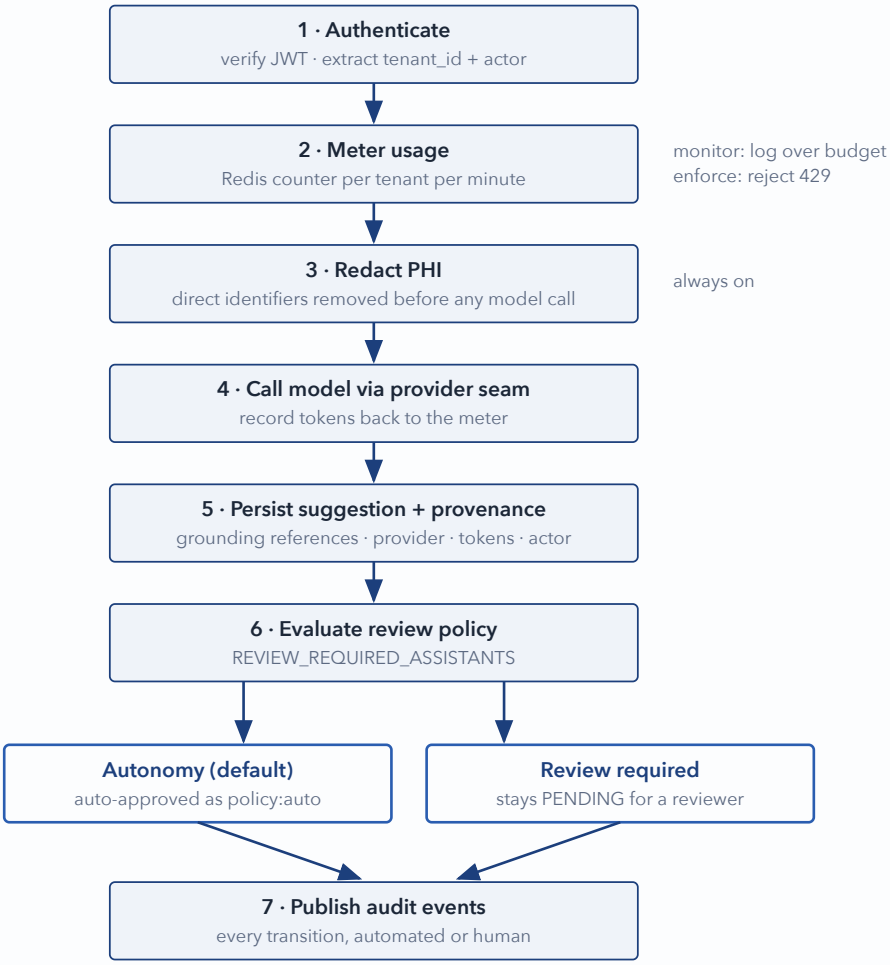


Figure 2 · Every request passes through the same plane in the same order

SECTION 03

Suggestion state machine

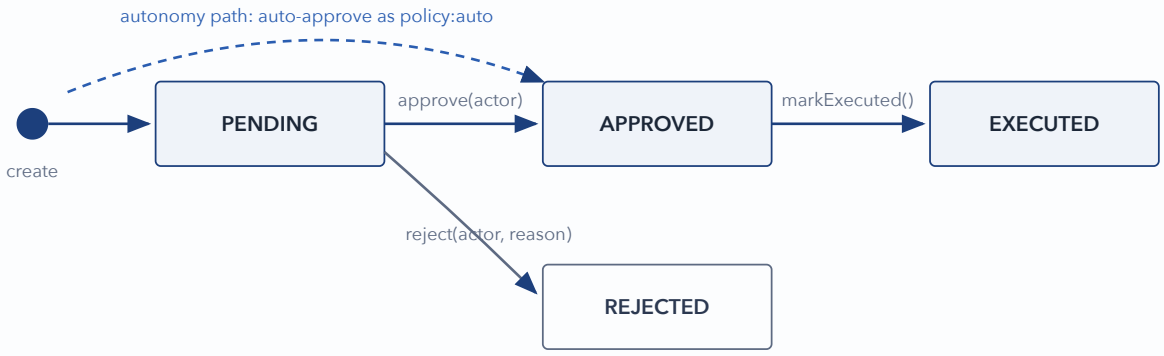


Figure 3 · Transitions enforced in the entity; decisions only from PENDING, execution only from APPROVED

SECTION 04

Deployment topology

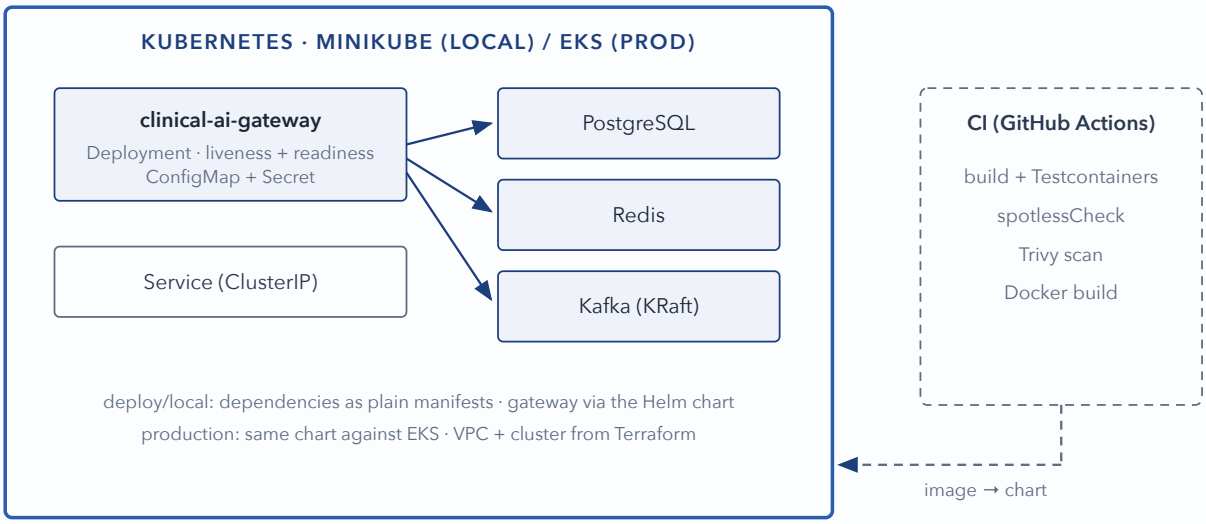


Figure 4 · One Helm chart serves the local minikube proof and the EKS target

SECTION 05

Decisions these diagrams encode

- Assistants never call the model directly; every request crosses the control plane, so governance is defined once and applied uniformly (ADR 0001).
- Review is a policy outcome inside the flow, not a separate approval system; the autonomous and reviewed paths share one state machine and one audit trail (ADR 0002).
- The audit trail leaves the request path through Kafka and lands in a separate read model, so new consumers attach without touching the producer (ADR 0003).
- PHI redaction sits at the gateway boundary, upstream of the provider seam, so no provider contract determines what data leaves the trust boundary (ADR 0004).
- The provider seam keeps the model layer swappable; the mock provider makes the whole plane runnable and testable with no cloud dependency (ADR 0005).
- Metering is observability first; enforcement is an opt-in mode, so a cost signal never becomes an availability incident by default (ADR 0006).
- One Helm chart deploys both targets, so the local cluster is a faithful rehearsal of production rather than a separate configuration.

END OF DOCUMENT · CAG-02